

PRATYUSH PADHY

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EDUCATION

University of California, Irvine

Irvine, CA

B.S., Computer Science, Specialization in Intelligent Systems, GPA: 3.86/4.00

Expected Graduation: 2028

- Dean's Honor List: Fall '25, Winter '26, Spring '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

East Palo Alto, CA

Incoming Software Development Engineer Intern, Open Data Analytics Engines (ODAE) Runtime

Sept. 2026 – Dec. 2026

- Joining the ODAE Runtime team, which owns the runtime for AWS's open-source analytics engines (**Spark**, **Trino**, **Flink**) behind **EMR**, **Athena**, and **Glue**, focused on large-scale distributed query execution and engine performance at cloud scale.

Tata Consultancy Services (TCS)

CMU Campus, Pittsburgh, PA

AI/ML Intern, TCS Medical Robotics Center

Jun. 2026 – Aug. 2026

- Benchmarked task-specific **RL** against a generalist **vision-language-action** policy (**SmolVLA** 450M, **LeRobot**) across **250+** episodes on an **AgileX Piper 6-DOF** arm; the VLA matched state-based RL (**85.3% vs. 81.5%**) using camera pixels alone.
- Extended **FluidLab** (ICLR '23 MPM fluid simulator) into a robotics platform: **URDF forward-kinematics** joint control (**0.0mm** error), analytic **SDF** liquid vessels on a **128³** voxel grid, an offscreen renderer, and **3** upstream engine-bug fixes.
- Diagnosed a **sparse-reward exploration collapse** and solved it by **action-space reparameterization** (no reward shaping, no demonstrations), training **PPO (Stable-Baselines3)** to **81.5% ± 3.3%** liquid-transfer success across 3 independent seeds.
- Fine-tuned SmolVLA via two-stage **imitation learning** (real-robot pretraining, then **120** PPO-generated demos) in **13 min**; budget-matched PPO trained on identical pixels scored **0%**, an **ablation** isolating pretrained vision + imitation.

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Designed a centralized storage layer, a storage-agnostic schema class over **MongoDB/SQLite** behind a **React** useContext provider, replacing **10+** scattered storage instances and batching persistence into a single write per user action.
- Built a course catalog with real-time retrieval of **1,200+** UCI courses, the headline feature of the funding demo that secured **\$5,500**; resolved **30+** production bugs across the stack, including a credits-calculation error affecting schedule planning.
- Redesigned the year planner from a rigid 4-year structure to support unlimited years with per-quarter course blocks.

PROJECTS

Chronicle | Python, FastAPI, Postgres/pgvector, Next.js | chronicles-weld.vercel.app

Jun. 2026 – Present

- Built a job aggregation platform ingesting **~26K** open roles from **600+** companies' ATS boards (**Greenhouse**, **Lever**, **Ashby**) via async fan-out with per-company fault isolation and idempotent **24–48h** refreshes that dedupe cross-source postings.
- Engineered hybrid search fusing **Postgres full-text (ts_rank_cd, GIN index)** with semantic ranking (int8 **ONNX** MiniLM, **pgvector** HNSW) via **RRF** at **p95 19ms** on a 20K-job corpus (local bench); a profile-vector recommender lifted **NDCG@10** 0.58→0.81, **recall@50** 0.71→0.96, and **MRR** 0.80→0.98 over the rule baseline (n=24 personas, bootstrap 95% CIs).
- Shipped **Google OAuth** accounts, kanban application tracking, saved-search email alerts, scheduled daily ingestion (**GitHub Actions** cron with an idempotent run-lock), a **119-test** CI suite gating every push, and a keep-warm probe against cold starts.

Forge | Python, PyTorch, Unsloth, TRL, vLLM, Ollama | forge-grpo.vercel.app

Jul. 2026 – Present

- Post-trained **Qwen2.5-1.5B** to reason via **GRPO** (RL from **verifiable rewards**, no human labels or reward model) with **Unsloth + TRL**, lifting **GSM8K pass@1** 58.8%→70.0% on 1,319 held-out problems in **86 min** at **3.64 GiB** peak VRAM.
- Designed the reward stack (correctness, format, graded tag-presence), diagnosing a zero-reward cold start that nulled GRPO's gradient; verified **100%** output-format compliance and no catastrophic forgetting on **ARC-Challenge** (69.5%→68.5%).
- Quantized to **GGUF Q4_K_M** (3.15× smaller) and served at **228 tok/s** (**Ollama**) and **674 tok/s** batched (**vLLM** multi-LoRA) behind a live base-vs-tuned playground running real streaming inference on a scale-to-zero GPU endpoint.

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV, dlib | face-pulse.netlify.app

Apr. 2026 – Jun. 2026

- Featured on the Handshake AI Showcase**, selected among thousands of student projects for national recognition; evolved from a facial-landmark detector into a comparative study of **4 architectures** (pixel MLP, landmark MLP, custom **CNN**, two-phase **VGG16**) on **FER2013** (35K+ images, 5 classes), the custom CNN leading at **64.5%** accuracy / **0.647** macro-F1.
- Built a full-stack app: live in-browser webcam detection (**React** + face-api.js) with real-time emotion probability charts and session analytics (**Chart.js**), plus a **FastAPI** + **TensorFlow** inference API serving the trained checkpoints (**Docker**, CI).

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS, Bash

ML & Concepts: reinforcement learning (PPO, GRPO), imitation learning & VLA fine-tuning, RAG & hybrid retrieval, LLM fine-tuning & quantization, embeddings & vector search, recommender systems, evaluation & ablations, NLP

Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, Transformers, Unsloth, TRL, vLLM, LeRobot, Stable-Baselines3, Gymnasium, OpenCV, NumPy, Pandas, Matplotlib, React, Next.js, FastAPI, SQLAlchemy, Tailwind CSS

Tools & Platforms: Git, GitHub Actions (CI), Docker, pytest, PostgreSQL, Neon, Ollama, ROS2, Gazebo, FluidLab, WSL2/CUDA, Vercel, Render, MongoDB, SQLite, Modal, Netlify